

Part 2 -- Classification: Predicting S&P 500 Stock Price Direction

1. Dataset Description

Source: S&P 500 Stocks (daily updated) -- Kaggle

This dataset contains daily OHLCV stock data for all S&P 500 companies. We used the same subset as Part 1 (top 50 companies by market cap, 2018-present).

Target variable: Binary -- 1 if next-day close > today's close (UP), 0 otherwise (DOWN). This transforms the regression problem into a binary classification of price direction.

Class balance: Approximately 52% UP days, 48% DOWN days -- near-balanced but F1 is still used as the primary metric since accuracy alone can be misleading for directional prediction.

Why this dataset? Predicting stock price direction is a canonical classification challenge in quantitative finance. The near-random nature of daily returns makes this a rigorous test of classification models.

2. Methodology

Feature Engineering

We engineered 13 technical indicator features from raw OHLCV data:

Feature Category	Features
Price-based	Daily Return, Daily Range (normalized), Price Spread
Momentum	5-day return, 20-day return, SMA ratio (5, 20)
Volume	Volume change, Volume ratio to 5-day MA
Volatility	5-day & 20-day rolling std of returns
Technical	RSI (14-day)
Categorical	Sector (encoded)

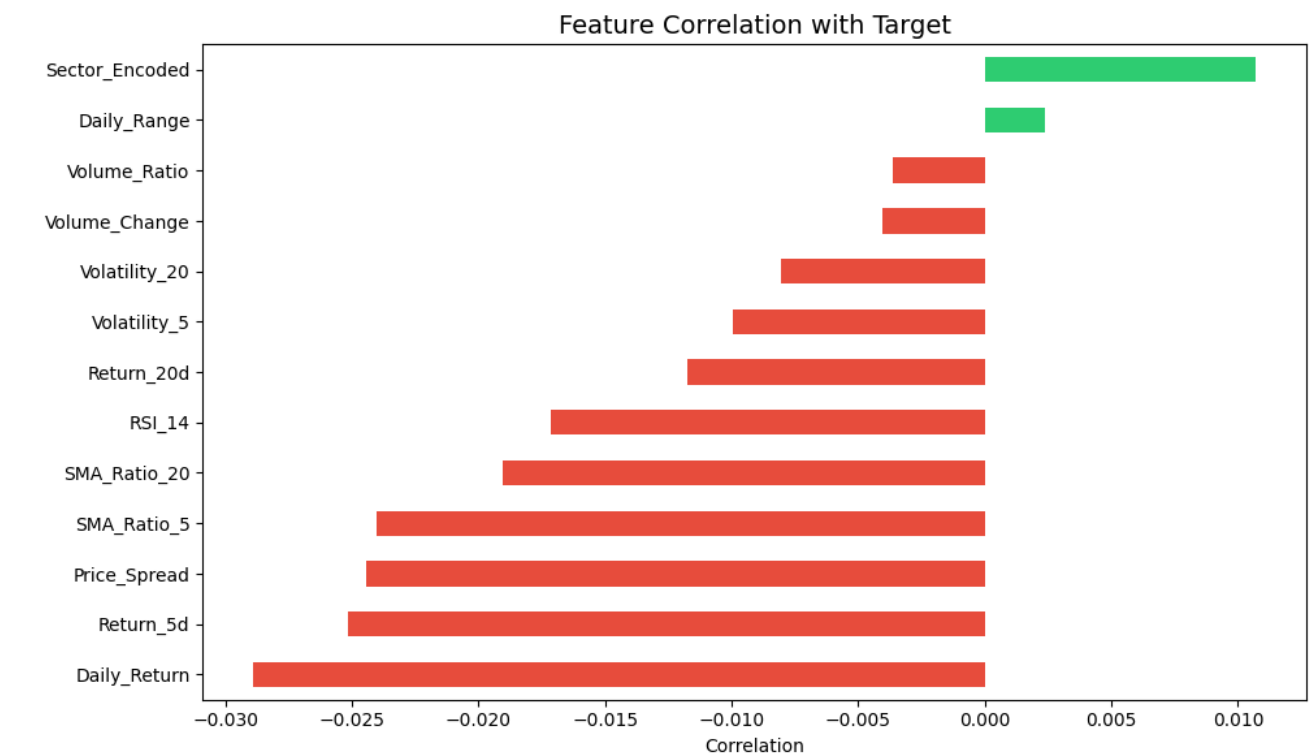


Figure 1: Feature correlation with target. Most features have weak correlation with direction, reflecting the inherent randomness of daily price movements.

Model Choices

#	Model	Key Configuration
1	Logistic Regression	max_iter=1000
2	SVC	Pipeline(StandardScaler -> RBF, C=1.0)
3	Decision Tree	max_depth=10
4	Random Forest	n_estimators=200, max_depth=15
5	Neural NetworkStandardScaler + 3 layers (64->32->16), sigmoid, binary_crossentropy	

SVC and Neural Network were wrapped with StandardScaler as mandated.

Hyperparameter Tuning

Random Forest Classifier was tuned via GridSearchCV (3-fold CV, scoring='f1'):

Parameter	Search Values	Best
n_estimators	100, 200, 300	300
max_depth	10, 15, 20, None	10
min_samples_split	2, 5, 10	10

3. Results

Model Comparison (sorted by F1 Score)

Model	Accuracy	F1 Score	Precision	Recall
Logistic Regression	0.5277	0.6829	0.5288	0.9636
Random Forest (Tuned)	0.5355	0.6778	0.5346	0.9259
SVC	0.5284	0.6730	0.5307	0.9196
Decision Tree	0.5293	0.6689	0.5320	0.9009
Random Forest (Default)	0.5319	0.6638	0.5346	0.8754
Neural Network (Keras)	0.5181	0.5730	0.5382	0.6126

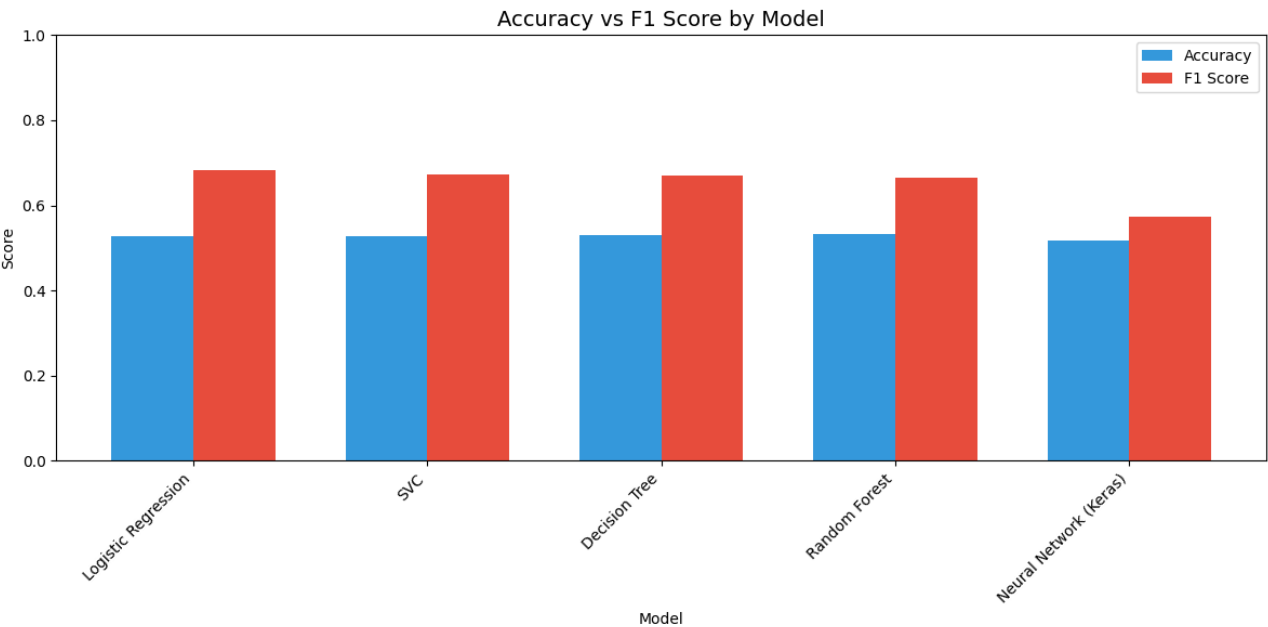


Figure 2: Accuracy vs F1 Score across all models. All models achieve similar accuracy (~53%), but F1 scores differ due to varying precision-recall trade-offs.

Key Findings

- All models achieve accuracies near 53%, only slightly above random (50%), which is expected -- daily stock movements are notoriously difficult to predict.
- Logistic Regression achieved the best F1 (0.6829) due to very high recall (0.9636), though with lower precision (0.5288).
- Tuning improved RF from F1=0.6638 to F1=0.6778 by regularizing with lower max_depth (10) and higher min_samples_split (10).
- The Decision Tree visualization (max_depth=3) reveals that Daily_Return and RSI are the primary split features.

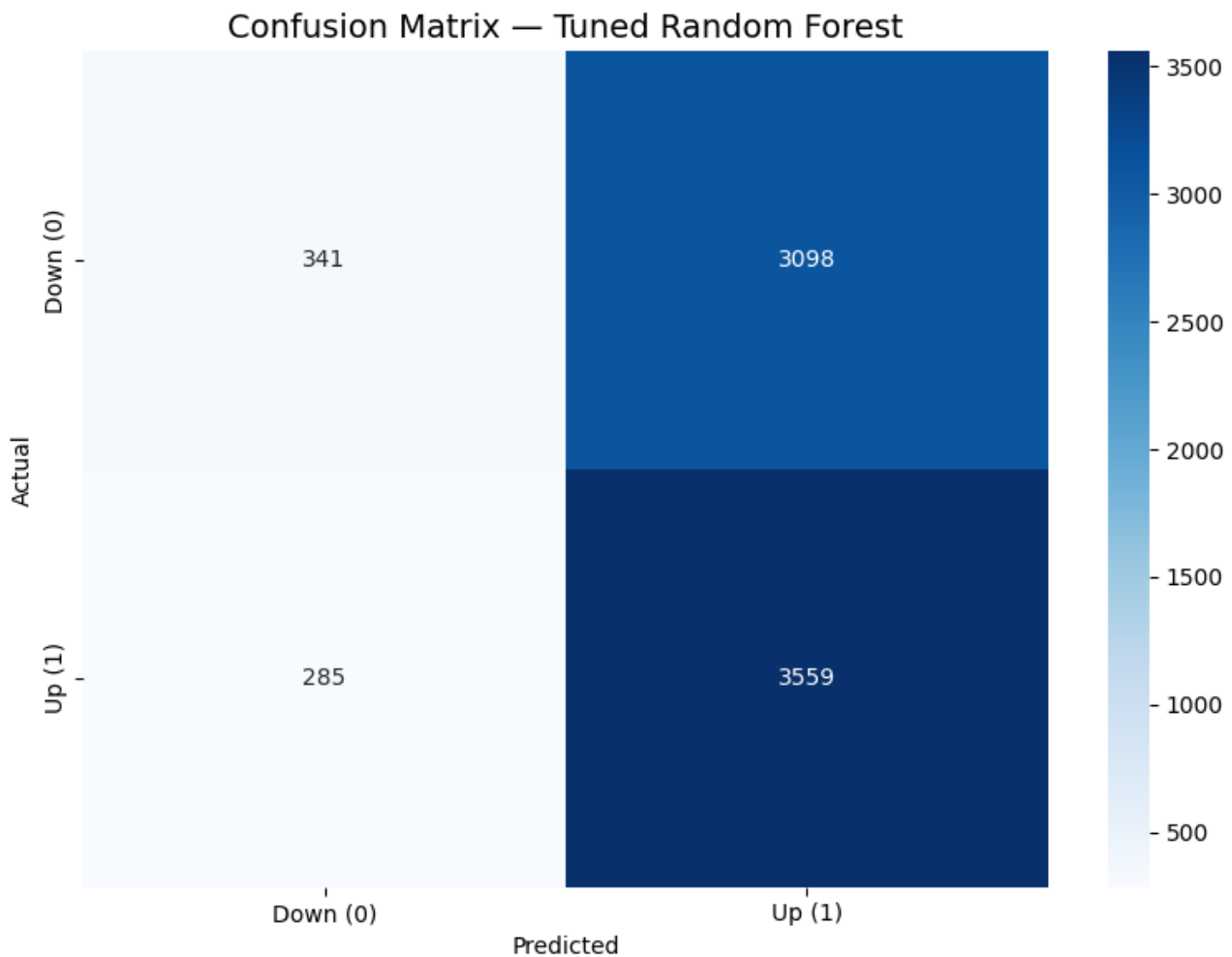


Figure 3: Confusion matrix for the tuned Random Forest -- shows slight bias toward predicting UP.

4. Conclusion

Recommended Model: Logistic Regression

Despite its simplicity, Logistic Regression achieved the highest F1 score (0.6829). Its high recall (0.9636) means it catches almost all UP days, though at the cost of many false positives. In practice, this suggests the model learns that the long-term market bias is upward.

The modest accuracy across all models confirms that daily stock direction is essentially a random walk with a slight upward drift -- a well-established result in financial literature. More sophisticated models (RF, NN) did not significantly outperform the logistic baseline, suggesting that the features capture limited predictive signal at the daily frequency.